Gate 10 — Husbandry Contribute: Django-backed Contact Form

> ***Status: Deferred to post-MVP (2026-04-19).*** *BA recommended descoping Gate 10 from the ABQ BioPark ECA Workshop (June 1–5, 2026) MVP. The Gate 09 mailto stub at* `frontend/app/contribute/husbandry/page.tsx` *remains the workshop-visible contribute surface. Reopen this gate as the first post-workshop backlog item, ideally paired with a small moderation-UI sub-gate (admin list filters, one-click accept/reject) so Aleksei isn't hand-copying contributions into* `SpeciesHusbandry` *forever. Full reasoning, scope reductions if plans change, and open questions: see* `docs/planning/business-analysis/gate-10-contribute-form-assessment-2026-04-19.md`*.*

Goal

Ship the "Contribute updates" CTA destination as a **Django-backed contact form** (locked Q4, 2026-04-18). Not mailto. Not Google Form. Submissions land in a Django model visible to Tier 5 in admin, preserving the platform audit trail and avoiding external dependencies. Design the data shape so it can graduate cleanly to a Tier 3+ moderated submission queue post-MVP without a model rewrite.

This is the lowest-priority of the three husbandry gates — if schedule slips, descope and ship Gate 09 with a placeholder CTA. The workshop demo does not require a live form, only visible intent.

Stories

- **Story 10.1** — As a Tier 1 public user, I want a form at `/contribute/husbandry/` that accepts my name, email, the species I am contributing about, a message, and optional citations/URLs, so that I can offer husbandry updates without a platform account.
- **Story 10.2** — As a Tier 1 user arriving from the husbandry page CTA (`?species={id}`), I want the species context pre-filled and locked (or easily overridable), so that I do not have to re-identify the species I just came from (AC-6 pre-fill clause).
- **Story 10.3** — As a Tier 5 admin, I want submitted contributions to land in a Django model (`HusbandryContribution`) and be visible in Django admin with filter by species and by status, so that I can triage pre-workshop volume manually.
- **Story 10.4** — As the platform operator, I want the submission model designed such that a future Tier 2/3 authenticated submission (linked to a `User` FK, moderation `status`, reviewer FK) fits without a destructive migration.
- **Story 10.5** — As a Tier 1 user submitting the form, I want spam mitigation (honeypot + per-IP rate limit) so the inbox does not get flooded pre-workshop.
- **Story 10.6** — As a Tier 5 admin, I want an email notification on new submissions (to a configurable address, defaulting to Aleksei's), so that I do not have to poll admin.

Scope Assessment

Story	Frontend	Backend	Full-Stack	Complexity
10.1			✓	M
10.2	✓			S
10.3		✓		S

10.4		✓		S
10.5		✓		S
10.6		✓		S

Full-stack gate. Smaller than Gates 08/09.

Data Model

New model `HusbandryContribution` (same `husbandry` app as Gate 08):

- `species` — `ForeignKey(Species, on_delete=SET_NULL, null=True, related_name='husbandry_contributions')` . Nullable so a submission is retained even if the species is later deleted, and to allow "species not listed" submissions (select-an-"other"-option on the form).
- `submitter_name` — `CharField(max_length=200)` , required.
- `submitter_email` — `EmailField` , required.
- `submitter_affiliation` — `CharField(max_length=300, blank=True)` .
- `submitter_user` — `ForeignKey(User, null=True, blank=True, on_delete=SET_NULL, related_name='husbandry_contributions')` . Null at MVP (anonymous submitters). **Field exists so the post-MVP Tier 2+ path slots in without migration pain.**
- `message` — `TextField` , required.
- `citations` — `TextField(blank=True)` — free text at MVP; post-MVP becomes a structured sources inline.
- `status` — `CharField` with choices `new` / `in_review` / `accepted` / `rejected` / `spam` , default `new` . **Field exists so the post-MVP moderation queue reuses it.**
- `reviewer` — `ForeignKey(User, null=True, blank=True, on_delete=SET_NULL, related_name='reviewed_husbandry_contributions')` . Null at MVP.
- `review_notes` — `TextField(blank=True)` .
- `submitter_ip` — `GenericIPAddressField(null=True, blank=True)` — recorded for rate-limit and spam triage. **Not** shown in admin list view by default (PII-adjacent). Retention: 90 days via a scheduled prune; prune is out-of-scope for this gate but the retention policy is documented here.
- `user_agent` — `CharField(max_length=500, blank=True)` .
- `created_at` — auto.
- `updated_at` — auto.
- Meta ordering `('-created_at',)` ; indexes on `status` and `species` .

API / View

- `POST /api/husbandry/contributions/` — public endpoint, honeypot field `website` (must be blank), rate-limited to 5 requests per IP per hour (use DRF throttling). Validates species exists (or accepts "other" with null FK). Returns 201 on success with no body content beyond `{"ok": true}` .
- `GET /api/husbandry/contributions/` — **not exposed** (Tier 5 reads via admin only at MVP).
- Email notification: on successful create, send plain-text email to `HUSBANDRY_CONTRIBUTION_NOTIFY` setting (default `alex.saunders@wildlifeprotectionsolutions.org`) with submission contents and a link to the admin change page. Use Django's email backend; SMTP config already in place.

Frontend Route

- **New page** `frontend/app/contribute/husbandry/page.tsx`.
- Form fields: Name, Email, Affiliation (optional), Species (select — pre- filled from `?species={id}` query param, editable; includes an "Other / not listed" option), Message, Citations (optional textarea).
- Honeypot field `website` rendered visually hidden (`sr-only` + tabindex=-1) — if filled, backend silently marks as spam.
- Client-side validation: required fields, email format.
- On submit: POST to the API; on 201 render a success state with the user's message echoed and a link back to the species profile. On error render an inline error with retry affordance. Do not wipe form state on error.
- Links back to `/species/[id]/` and (if species known) to `/species/[id]/husbandry/` in the confirmation state.

Acceptance Criteria

AC-10.1 — Form pre-fills species from query param

Given a user clicks "Contribute updates" on `/species/[id=X]/husbandry/` **When** they arrive at `/contribute/husbandry/?species={X}` **Then** the Species selector is pre-populated with species X and the form is otherwise empty.

AC-10.2 — Species pre-fill is editable

Given the form has pre-filled species X from the query param **When** the user opens the species selector **Then** they can change to a different species or to the "Other / not listed" option; the form does not force them to submit about species X.

AC-10.3 — Valid submission creates a record

Given a user fills name, email, species (X), and message **When** they submit **Then** the API returns 201, a `HusbandryContribution` row is created with `status='new'`, `species=X`, `submitter_user=NULL`, `submitter_ip` populated, and the confirmation state renders on the client.

AC-10.4 — Missing required fields are rejected

Given a user submits the form with an empty `message` **When** the server validates **Then** the response is 400 with a field-level error on `message` and no row is created.

AC-10.5 — Honeypot silently rejects spam

Given a bot submits the form with the hidden `website` field filled **When** the server receives the request **Then** the response is 201 (to avoid signaling the honeypot exists), a row **may** be created with `status='spam'`, and no notification email is sent.

AC-10.6 — Rate limit enforced

Given a client has submitted 5 successful requests from the same IP in the past hour **When** they submit a 6th **Then** the response is 429 with a `Retry-After` header and no row is created.

AC-10.7 — Admin sees submissions

Given a submission exists **When** a Tier 5 admin opens the `HusbandryContribution` admin list view **Then** the row is visible with species, submitter_name, submitter_email, created_at, and status columns; filters for `status` and `species` are available; the submitter_ip column is not shown in the list view but is readable on the change form.

AC-10.8 — Notification email fires

Given a successful (non-spam) submission **When** the transaction commits **Then** an email is sent to `settings.HUSBANDRY_CONTRIBUTION_NOTIFY` containing submitter name, email, species name, message, citations, and a link to the admin change page.

AC-10.9 — Model supports future Tier 2+ moderation without migration

Given the `HusbandryContribution` model as shipped **When** a future gate adds a Tier 2+ authenticated submission pipeline **Then** the existing `submitter_user`, `status`, `reviewer`, `review_notes` fields accommodate the new flow with no destructive migration — the only change is flipping `submitter_user` to non-null for authenticated paths and adding new `status` transitions.

AC-10.10 — "Species not listed" path works

Given a user selects "Other / not listed" and fills the form **When** they submit **Then** the record is created with `species=NULL` and the message body is the sole species identifier for triage.

Out of Scope

- Tier 2+ authenticated submission UI (post-MVP).
- Inline structured `HusbandrySource` citations on the form (post-MVP; free-text `citations` field holds the shape).
- Moderation workflow beyond manual admin status changes (post-MVP).
- Automatic merge of accepted contributions into `SpeciesHusbandry` records (post-MVP — every accepted contribution still requires an admin to author the change on the record).
- CAPTCHA (honeypot + rate-limit is sufficient at MVP volume; revisit if spam lands).
- File attachments (post-MVP).
- IP retention / pruning automation (policy documented here; implementation post-MVP).

Dependencies

- Gate 08's `husbandry` Django app exists.
- Gate 09 is shipping the CTA that links here. If Gate 10 ships first, the route exists but no one reaches it organically — acceptable.
- Django email backend configured in staging + production (already in place).
- DRF throttling (already configured for other endpoints).

Sequencing / Deadline Note (June 1, 2026)

Lowest priority of the three husbandry gates. Target merge: **2026-05-28**. If Gate 08 or Gate 09 slip, descope this gate and ship Gate 09 with the CTA pointing to a static "Email us at {address}" placeholder. The SHOAL conversation at the workshop is about the **frame and the invitation to collaborate** (per BA §6) — a live form is credibility- positive but not credibility-critical. An obvious placeholder ("form coming soon — email us") is acceptable; a form that silently 500s is not.

Test Writer Guidance

At this gate, the test writer should verify:

- Happy-path submission creates a row and fires an email.
- Honeypot rejects silently without email.
- Rate limit kicks in at the 6th request from one IP within an hour and resets on the rolling window.
- Missing required fields produce 400 with field-level errors.
- Query-param pre-fill arrives at the client page and is editable.
- "Other / not listed" option records `species=NULL`.
- Admin list view respects the filter + column spec; `submitter_ip` is hidden from the list view.
- **Adversarial:** SQL injection attempts in `message` / `citations` are escaped (covered by Django ORM — assert rather than actively exploit).
- **Adversarial:** submitting with `submitter_user` in the POST body is ignored (the field is server-assigned from request auth state; at MVP always NULL for anonymous POSTs).
- **Adversarial:** repeated requests that would exceed the rate limit return 429 and do **not** create half-submitted rows.
- **Adversarial:** attempting `GET /api/husbandry/contributions/` as any tier returns 405/403 — listing submissions is admin-only.
- Email notification contains no sensitive platform data beyond what the submitter typed plus the species name.

Risks and Open Questions

- **Notification email deliverability** depends on production SMTP config. Verify a test submission lands in Aleksei's inbox in staging before cut-over to prod.
- **Spam volume pre-workshop.** The ECA Workshop will publicize the platform. If spam spikes, the honeypot + 5/hour/IP throttle may be insufficient; be prepared to add CAPTCHA post-workshop.
- **IP retention.** 90-day retention policy is documented but not automated. If GDPR-style data requests arrive, manual deletion via admin is acceptable at MVP volume.
- **Pre-filled species trustworthiness.** The `?species={id}` param is user-controllable; the server validates the FK on submit. A malicious actor cannot attribute a submission to the wrong species without the submitter noticing in the UI.